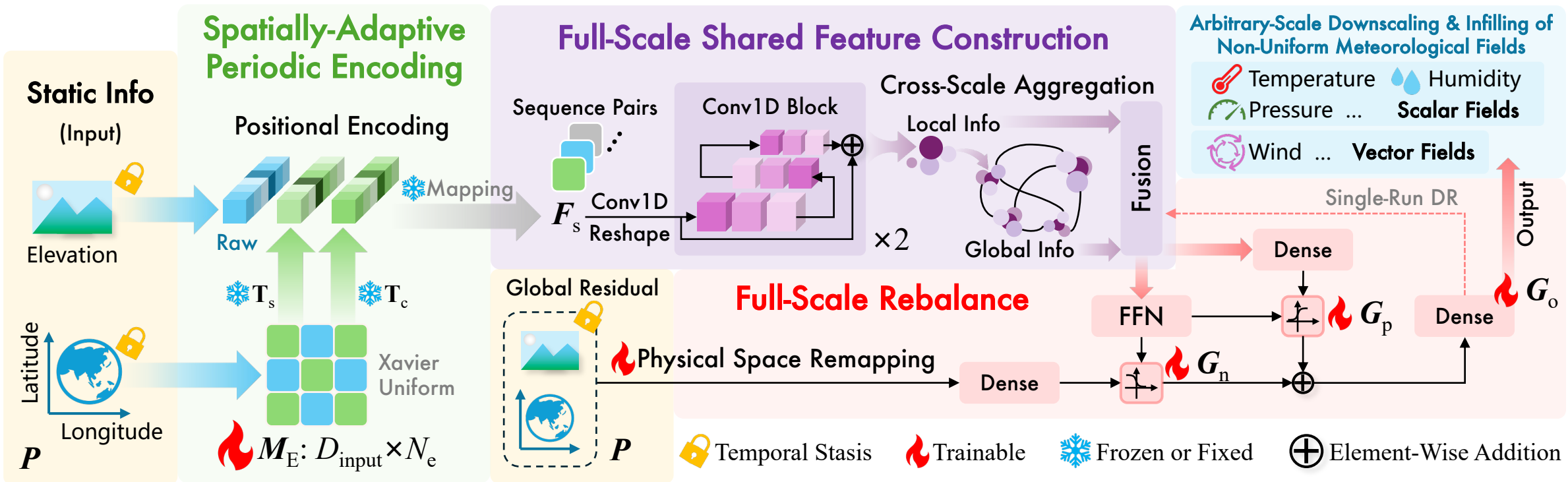
Static Info
(Input)
Elevation
Latitude
Longitude
P
Spatially-Adaptive Periodic Encoding
Positional Encoding
Raw
Mapping
Ts
Tc
Xavier Uniform
ME: Dinput×Ne
Full-Scale Shared Feature Construction
Sequence Pairs
Fs
Conv1D
Reshape
Conv1D Block
×2
Cross-Scale Aggregation
Local Info
Global Info
Fusion
Arbitrary-Scale Downscaling & Infilling of Non-Uniform Meteorological Fields
Temperature
Humidity
Pressure ...
Scalar Fields
Wind ...
Vector Fields
Single-Run DR
Output
Dense
Global Residual
Full-Scale Rebalance
FFN
Gp
Go
Gn
Physical Space Remapping
Dense
Dense
P
Temporal Stasis
Trainable
Frozen or Fixed
Element-Wise Addition